

Midhat Ratib Khan | Data Scientist

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Education

North South University

B.Sc. in Computer Science and Engineering

Specialization: Artificial Intelligence and Machine Learning Track

Dhaka

2021–2025

Projects

DataBrief — Automated Data Storytelling Platform

Live • GitHub

- Built and deployed a production SaaS that transforms raw CSV/Excel datasets into narrative analytical reports (PDF, DOCX, PPTX), combining a 6-type statistical analysis engine (descriptive statistics, trend detection, outlier identification, correlation, distribution profiling, ranking) with LLM-generated narration.
- Designed a question-aware analysis pipeline: a lightweight (~100-token) LLM call identifies question-relevant columns before statistical analysis runs, plus an insight-scoring system that ranks findings by importance to structure each report's story arc.
- Architected asynchronous infrastructure with FastAPI, Celery/Redis workers, and PostgreSQL with multi-tenant row-level security; exports stored in Cloudflare R2 with pre-signed URLs, fallback template narration for LLM-API outages, Dockerized deployment across Railway and Vercel.

FinRatio — Corporate Leverage vs. Profitability Analysis

- Framed and tested the business hypothesis “does more debt mean more profit”: computed leverage (debt-to-equity, debt-to-assets) and profitability ratios (net profit margin, ROE, ROA) across multiple companies and industries.
- Conducted end-to-end EDA and Pearson correlation analysis (pandas, NumPy, SciPy); found a null-to-slightly-negative relationship between leverage and profit margins, contradicting the assumption that debt amplifies returns.
- Communicated the counterintuitive finding to non-technical audiences via scatter plots, distributions, and correlation heatmaps (matplotlib, seaborn) in a publicly deployed narrative report.

PorteHobe AI — Multi-Agent LLM Tutoring System (Capstone)

- Co-built an autonomous teaching platform using a Router → Specialist → Judge multi-agent architecture across small language models (Qwen, Mathstral, Gemma), orchestrated with LangGraph and FastAPI.
- Led the empirical evaluation pipeline: benchmarked mathematical reasoning on GSM8K and MathOdyssey against state-of-the-art LLM baselines to quantify routing gains in accuracy and efficiency.

SmartRoute-LLM:

- Efficient Model Routing and Prompt Optimization for Math and Code Reasoning using Small Language Models — Co-author, IEEE.

Professional Experience

Innovation for Wellbeing Foundation

Technical Consultant (Web)

Dhaka

Oct 2024–Mar 2025

- Led development and consulting for “Hearth Summit Bangladesh,” an international brand launch: built and deployed the CMS-based web platform and supporting infrastructure end to end.

IEEE NSU Student Chapter

Web Development Team

Dhaka

Mar 2024–Nov 2024

- Built and deployed a Django-based email automation service powering the chapter's institutional outreach.

Technical Skills

Data Science & Analysis: pandas, NumPy, SciPy, scikit-learn, statistical analysis (correlation, distributions, outlier detection), EDA, matplotlib, seaborn, Jupyter

Languages & Databases: Python, SQL (PostgreSQL, MySQL), TypeScript

LLM & AI Engineering: LangChain, LangGraph, RAG, model routing, prompt engineering, benchmark design (GSM8K, MathOdyssey), Groq, Ollama

Engineering & Deployment: FastAPI, Celery, Redis, Next.js, Docker, Git/GitHub, Railway, Vercel, Supabase, Linux

Spoken Languages: English (professional working proficiency) • Bangla (native)